JULIA MARCOLAN

PhD Researcher in Computational Physics | Nuclear Magnetic Resonance (NMR) | Magnetic Resonance Imaging (MRI) | Pulse sequence Design

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Complete publication list: Lattes Curriculum

RESEARCH PROFILE

PhD researcher in Computational Physics at the São Carlos Institute of Physics (IFSC), University of São Paulo (USP), specializing in magnetic resonance pulse sequence design and development. Research interests include low- and mid-field MRI, time-domain NMR, low-field NMR spectroscopy, and the development of novel acquisition methods for unconventional and challenging experimental conditions. Experienced in developing methods for low-field J-spectroscopy and in designing and evaluating MRI techniques for highly inhomogeneous magnetic fields, including adiabatic pulse sequences and quantitative B_1 and T_1 mapping.

EDUCATION

PhD in Computational Physics
University of São Paulo (USP), São Carlos, Brazil
2023-Present

MSc in Physics / Computational Physics
University of São Paulo (USP), São Carlos, Brazil
2021-2023

BSc in Computational Physics
Federal Fluminense University (UFF), Brazil
2014-2019

RESEARCH EXPERIENCE

PhD Researcher — Magnetic Resonance Imaging and Nuclear Magnetic Resonance
University of São Paulo (USP), São Carlos, Brazil
2023-Present

- Development and implementation of MRI pulse sequences.

- Signal processing, quantitative analysis, and scientific visualization using Python.
- Developing methods for low-field J-spectroscopy.

Visiting Researcher / Research Collaboration
Center for Magnetic Resonance Research (CMRR), University of Minnesota, USA
2026

- Research on MRI methods for highly inhomogeneous magnetic fields.
- Investigation of pulse sequences using adiabatic pulses.
- Investigation of B_1 mapping methods using adiabatic frequency-modulated RF pulses.
- Development of methods for B_1 and T_1 mapping in highly inhomogeneous magnetic fields.

RESEARCH INTERESTS

- Pulse sequence design
- Magnetic Resonance Imaging (MRI)
- Time Domain Nuclear Magnetic Resonance (TD-NMR)
- Low-Field spectroscopy
- B_1 and T_1 mapping
- Adiabatic RF pulses
- Computational physics and numerical methods

SKILLS

PERSONAL SKILLS

- Strong interpersonal communication.
- Organization and attention to detail.
- Proactive problem-solving.
- Analytical skills and critical thinking.
- Ability to work effectively in a team.
- Creativity in solving technical problems.

TECHNICAL SKILLS

- Python for scientific computing.
- Programming in C and Fortran.
- Development of NMR/MRI methods.
- NMR signal and image processing.
- Technical and scientific documentation.

CONFERENCE ABSTRACTS AND PRESENTATIONS

ORAL PRESENTATIONS

1. **67th Experimental Nuclear Magnetic Resonance Conference (ENC 2026)**. *Towards Quantification of J-Coupled Molecules in Aqueous Solutions with J-Spectroscopy at Low and Inhomogeneous Magnetic Field*. Pacific Grove, California, USA, April 2026.
2. **16th International Conference on the Applications of Magnetic Resonance in Food Science (MR Food)**. *Magnetic Resonance Imaging for Short Relaxation Times Applied to Seed Evaluation*. 2024.

POSTER PRESENTATIONS

1. **1st Annual Meeting of the Brazilian Chapter of the International Society for Magnetic Resonance in Medicine (BC-ISMRM)**. *Magnetic Resonance Imaging with Ultrashort Relaxation Times Applied to Seed Evaluation*. 2025.
2. **XVIII Brazilian Magnetic Resonance Meeting (*Jornada Brasileira de Ressonância Magnética*)**. *Strategies for Obtaining J Spectra at Low Field*. 2024.
3. **XVII Brazilian Magnetic Resonance Meeting (*Jornada Brasileira de Ressonância Magnética*)**. *Calibration and Measurement Methodology Applied to the CIERMag Digital Magnetic Resonance Spectrometer (DMRS)*. 2022.
4. **2nd Graduate Research Meeting of the University of São Paulo (*Encontro da Pós-Graduação da USP*)**. *Development of a Single-Voxel Stimulated Echo Acquisition Mode (STEAM) Spectroscopy Method Compatible with the CIERMag Digital Spectrometer*. 2021.

AWARDS AND FELLOWSHIPS

2026

- Awarded the **ENC Student Travel Stipend 2026**, Experimental Nuclear Magnetic Resonance Conference (ENC), USA.

2025

- **PDSE/CAPES Scholarship** - Institutional Doctoral Sandwich Program Abroad, University of Minnesota, USA.

2024

- **2nd Place for Best Poster Presentation**, XVIII Brazilian Magnetic Resonance Meeting, *Magnetic Resonance in Chemistry*.
- **Finalist**, Yvonne Primerano Mascarenhas Award for Academic Excellence (PhD category), during the Integrated Week of the São Carlos Institute of Physics (SIFSC).

2023

- **PhD Scholarship**, CAPES (Coordination for the Improvement of Higher Education Personnel), 2023-Present.
- **Honorable Mention**, Yvonne Primerano Mascarenhas Award for Academic Excellence (PhD category), during the Integrated Week of the São Carlos Institute of Physics (SIFSC).

2022

- **Best Poster Presentation**, XVII Brazilian Magnetic Resonance Meeting, *Magnetic Resonance in Chemistry*.

2021

- **Master's Scholarship**, CNPq (National Council for Scientific and Technological Development), 2021-2023.

TEACHING AND MENTORING

Undergraduate Research Supervision

Maria Vitória Lima da Silva | 2024–2026

Project: Implementation of 2D Relaxometry Methods by Magnetic Resonance in the Digital Magnetic Resonance Spectrometer (DMRS).

Caio Fernandes Marques | 2024–2025

Project: Implementation of the Inverse Laplace Transform as a Tool for the Digital Magnetic Resonance Spectrometer (DMRS).

Outcome: Finalist project for the Yvonne Primerano Mascarenhas Award for Academic Excellence (Undergraduate Research category).

Teaching Experience

Teaching Assistant — Introduction to Computational Physics
São Carlos Institute of Physics (IFSC), University of São Paulo (USP), São Carlos, Brazil
August 2025 – December 2025

- Assisted in teaching the undergraduate course *Introduction to Computational Physics*.
- Supervisor: Prof. Dr. Francisco Castilho Alcaraz.

Teaching Assistant — Physics I, Teaching Improvement Program (PAE)
São Carlos Institute of Physics (IFSC), University of São Paulo (USP), São Carlos, Brazil
February 2025 – June 2025

- Participated in the University of São Paulo's Teaching Improvement Program (PAE), assisting with the undergraduate course *Physics I*.